

Quantum IS Notes 2

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2 One Quantum Bit

2.2 Superposition

2.2.1 Zero or One

Qubit can take two values:

$$|0\rangle \text{ or } |1\rangle$$

This is in "*ket*" format.

The bloch sphere is a way to visualize these distinct states.
 $|0\rangle$ is the north pole of the bloch sphere.
 $|1\rangle$ is the south pole of the bloch sphere.

2.2.2 Superposition

A superposition is a combination of $|0\rangle$ and $|1\rangle$.

For example:

Math	Coords
$\frac{1}{\sqrt{2}}(0\rangle + 1\rangle)$	$(1, 0, 0)$
$\frac{1}{\sqrt{2}}(0\rangle - 1\rangle)$	$(-1, 0, 0)$
$\frac{1}{\sqrt{2}}(0\rangle + i 1\rangle)$	$(0, -1, 0)$
$\frac{1}{\sqrt{2}}(0\rangle - i 1\rangle)$	$(0, 1, 0)$

These are commonly represented as:

$$\begin{aligned} |+\rangle &= \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \\ |-\rangle &= \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) \end{aligned}$$

$$|i\rangle = \frac{1}{\sqrt{2}}(|0\rangle + i|1\rangle)$$

$$|-i\rangle = \frac{1}{\sqrt{2}}(|0\rangle - i|1\rangle)$$

Or another way to represent it is:

